

Feature Selection Techniques in Machine Learning

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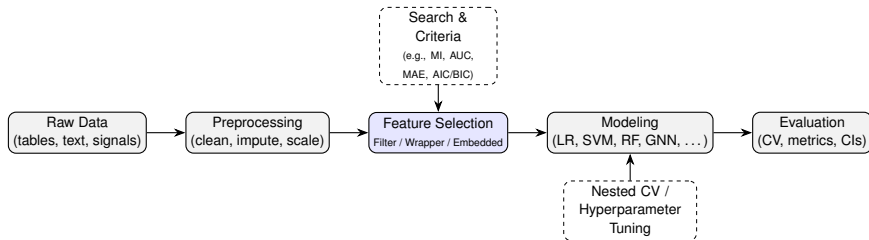
Outline

- 1 Introduction
- 2 Why Feature Selection?
- 3 Taxonomy of Methods
- 4 Worked Examples
- 5 Practical Guidance
- 6 Conclusion

What is Feature Selection?

- Selecting a subset of relevant variables for building predictive models.
- Goals: improve generalization, reduce overfitting, shorten training time, and enhance interpretability.
- **Distinction:** Feature *selection* (subset of existing variables) vs. Feature *extraction* (new transformed variables, e.g., PCA).

End-to-End Pipeline (with Feature Selection)



Motivation

- **Statistical**: mitigate high variance, reduce collinearity.
- **Computational**: faster training/inference; cheaper storage.
- **Scientific**: interpretability, causality hypotheses, measurement cost.
- **Operational**: robust deployment with fewer sensors/features.

Three Families

Filter

- Model-agnostic statistics: correlation, χ^2 , ANOVA F, mutual information (MI).
- Fast; low overfitting risk; may ignore interactions.

Wrapper

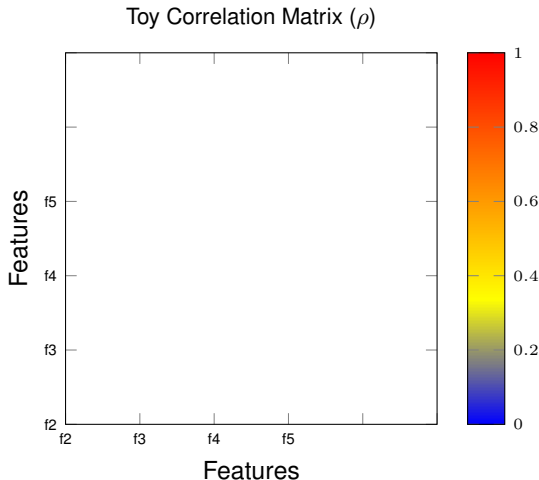
- Use a predictive model to score subsets: forward, backward, bidirectional, RFE.
- High accuracy potential; computationally intensive.

Embedded

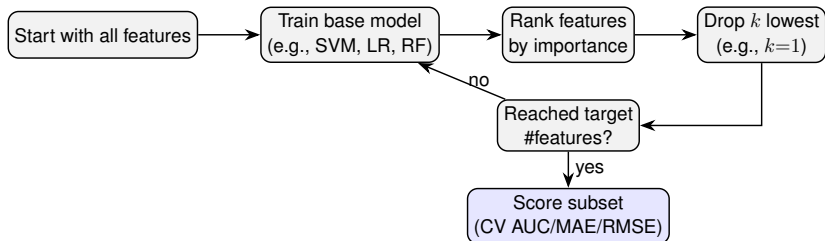
- Selection during training: Lasso/L1, trees/ensembles (Gini/MDI), elastic net.
- Balanced cost and performance; model-dependent.

Filter Example: Correlation Screening (Heatmap)

- Remove highly correlated features (e.g., $|\rho| > 0.9$) to stabilize linear models.
- Keep the one with higher univariate predictive value (e.g., MI with target).



Wrapper Example: Recursive Feature Elimination (RFE)



Tip: Use *nested CV* to avoid optimistic bias when tuning the number of features.

Embedded Examples: L1 & Tree-Based

Lasso / L1-regularized LR

- Adds $\lambda \|\beta\|_1$ encouraging sparsity.
- Coefficients of irrelevant features shrink to zero.
- Works well when a few features are truly predictive.

Elastic Net

- Mix of L1 and L2 helps with correlated groups.

Trees / Ensembles

- Feature importance via impurity decrease or permutation importance.
- Captures non-linearities & interactions.
- Beware: impurity-based importances can be biased to high-cardinality features.

Method Comparison

	Filter	Wrapper	Embedded
Model dependence	No	Yes	Yes
Speed	High	Low–Medium	Medium
Overfitting risk	Low	High (if not nested)	Medium
Interactions	Limited	Captured	Captured (model-specific)
Typical use	Quick screening	Small/medium p	High-dimensional sparse

Mini Case Study: Pima Diabetes (Logistic Regression)

Setup: 8 features, binary target.

- 1 *Filter:* Remove one of any pair with $|\rho| > 0.9$; rank by MI with target.
- 2 *Wrapper:* RFE with LR (liblinear), select top- k via inner CV.
- 3 *Embedded:* L1-penalized LR; choose λ via inner CV.
- 4 *Outer CV:* Compare test AUC/accuracy, report mean $\pm$ std and 95% CIs.

Outcome (illustrative): L1 selected 4 features, reducing variance and slightly improving AUC vs. full model.

Code Sketch: Filter $\rightarrow$ RFE $\rightarrow$ LR (scikit-learn)

```
pipe = Pipeline([ ("scale", StandardScaler()), ("clf",  
LogisticRegression(penalty="l2", solver="liblinear")) ])  
RFE over the classifier rfe =  
RFE(estimator=pipe.named_steps["clf"], n_features_to_select = 5, step = 1)  
Nested in a new pipeline for CV from sklearn.pipeline import  
make_pipeline  
make_pipeline(StandardScaler(), rfe, LogisticRegression(penalty =  
"l2", solver = "liblinear"))  
cv = StratifiedKFold(n_splits = 5, shuffle = True, random_state =  
42)  
scores = cross_val_score(nested, X, y, cv = cv, scoring =  
"roc_auc")  
print(scores.mean(), scores.std())
```

Practical Tips

- Always apply selection *inside* CV folds to prevent leakage.
- Prefer permutation importance over impurity for trees.
- For many correlated features, use Elastic Net or group-wise selection.
- Consider cost-aware selection when some features are expensive to acquire.
- Report stability of selected subsets (e.g., Jaccard across folds).

Summary

- Filters for quick, robust screening.
- Wrappers (RFE) for targeted subset optimization (use nested CV).
- Embedded (L1, trees) for efficient, model-informed sparsity.
- Hybrid pipelines often perform best in practice.

Thank You!

Questions?